

An Ultra-Low-Power IoT ToF Sensor Node

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1 - Introduction

My GCSE Technology project was a bin-mounted object counter, built around an IR LED, photodiode and 9V battery. It worked but the prototype was fragile, bulky and power-hungry, unsuitable for commercial production. This project is the result of wanting to develop the concept properly. An almost product-grade redesign that also served as my foundation to learning a modern Bluetooth SoC platform. The project was primarily funded by my Arkwright Scholarship, with PCB manufacturing sponsored by PCBWay. Development began in August 2025, with v1.0 releasing during April 2026.

The project is called AZ Sensor Suite (AZSS).

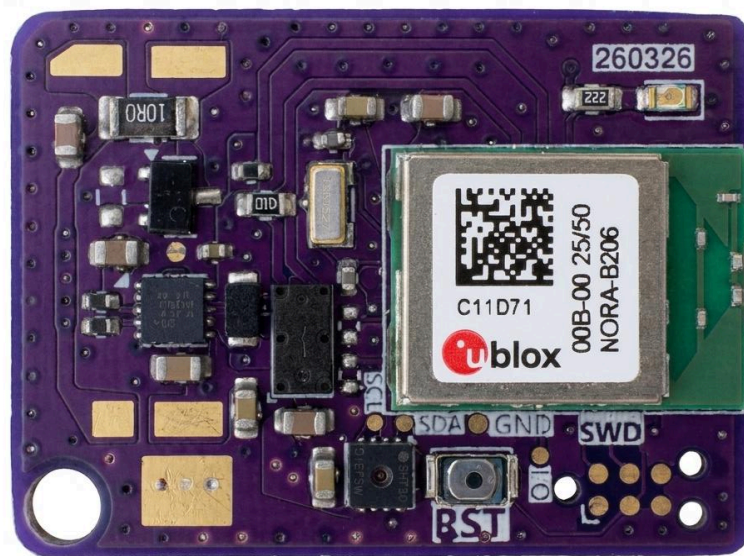


Fig 1.0.0.1 - Assembled PCB v1.11 (30.2 x 22.1 mm)

2 - The Problem to be Solved

There does not exist a low cost (Spec 2.1), small form factor (Spec 2.2), ultra-low power (Spec 2.3), IoT enabled (Spec 2.4), unified development platform for distance-measurement and presence detection applications. This means it is difficult to leverage ToF sensing in heavily constrained environments.

Example applications include:

- **Fill-level monitoring** for storage containers
- **Presence-triggered network input** (e.g., for smart home automation)
- Unidirectional **throughput** and **object counting**
- Simple **tamper detection**

The device should operate autonomously via direct wireless connectivity or be able to integrate into broader smart-home or industrial networks (Spec 2.5), and should be highly accurate (Spec 2.6) so that obtained data is reliable and usable for analysis.

3 - Research

3.1 - Customer-base and usage model

The project is primarily an open-hardware platform, with user responsibility for their own application firmware. To improve ease of development, a reference firmware implementation is provided for unidirectional object counting with Bluetooth connectivity.

The project caters to two market segments

- **Conventional purchase model** - The user programs the device independently. These users retain total control over the project. They prioritise reliability, repairability and ease of programming. (Spec 3.1)
- **Hardware-as-a-Service (HaaS) model** - The user receives a turnkey solution that does not require custom firmware development. Under this rental model, the user does not have to manage installation and maintenance, over a typical two-year lifecycle. (Spec 3.2)

This report will focus on users that fall under the conventional purchase model.

3.2 - Market solutions

As established in the problem analysis, no commercial solutions currently address this specific niche. The closest market analogues are official sensor evaluation breakout boards. While these manufacturer-provided modules break out the Time-of-Flight (ToF) sensor pins for initial firmware validation, they carry significant limitations for deployment:

- **Lack of integration** - They only feature the sensor, essential passives and a pin header. They lack onboard processing, power management, or wireless connectivity.
- **Poor form factor** - They require external processors to be connected via jumper wires, making them bulky and structurally unsuitable for permanent and/or mobile applications.
- **High unit cost** - They are produced in low volumes for development purposes, hence have a premium cost. A single breakout board costs more than the total BoM cost of the AZSS.

Breakout boards were used extensively during the initial phases of this project to validate firmware and characterise the sensor. However, this development process highlighted their limitations; the resulting breadboard setups were highly fragile, expensive, and unsuitable for any deployment.

3.3 - Hardware research

3.3.1 - Power Source and Topology

To maximise its deployability, the project requires an ultra small form factor, which implies an ultra-low-power budget. A primary CR2032 coin cell was selected over secondary lithium alternatives due to widespread availability and the elimination of space-demanding charging circuitry.

A CR2032 cell features a nominal voltage of 3.0 V with a usable range of roughly 2.0 V to 3.2 V. Due to the system's target operating voltage being 2.8 V, a buck-boost topology was required to efficiently utilise the full voltage range. The TI TPS63900 was chosen as a buck-boost converter with low quiescent current of 75 nA and high theoretical efficiency of approximately 88% under coin cell power.

3.3.2 - Distance Sensing Technology

Two ToF technologies were evaluated:

Selection Criteria	Infrared (IR) Laser ToF	Ultrasonic ToF
Key Advantages	• Low cost • Simple API integration • Narrow FoV without optics	• Lower active current draw • Optical material agnosticity • Higher sampling rate
Key Limitations	• Higher current draw • Lower maximum operating range	• High component cost • Requires mechanical horn to narrow FoV

Fig 3.3.2.1 -
Comparison of infrared and ultrasonic ToF sensing

IR laser sensing was selected to minimise hardware complexity and cost per unit. Specifically, the **STMicroelectronics VL53L4CD** was chosen as it includes an autonomous, ultra-low-power presence detection API that reduces current consumption to as little as 55 μA [1], and operates on a relatively low supply voltage of 2.8 V.

3.3.3 - System on Chip (SoC) Selection

The core SoC architecture requires capabilities to use a wireless protocol capable of integrating with smart-home networks without exceeding the strict power budget imposed by a coin cell. Wi-Fi connectivity was ruled out immediately due to its high peak current, which would rapidly deplete or brown out a CR2032 cell. Consequently, low-power personal wireless area network (LP-WPAN) protocols, specifically Bluetooth Low Energy (BLE), Thread, and Zigbee, were identified as viable IoT protocols for home and network integration.

To fulfil these requirements, the **Nordic Semiconductor nRF54L15** was selected. As a state of the art SoC manufactured on TSMC's 22ULL (22 nm, ultra low leakage) process node, it delivers double the processing performance and triple the energy efficiency of previous generation hardware. It supports multi-protocol stacks, features an ultra low sleep current (as low as 0.7 μA [2]), and offers a robust SDK ecosystem, with free training material provided by Nordic Semiconductor.

Crucially, its operating voltage matches the sensor rail, enabling the entire system to run off a single power supply and eliminating the space, cost, and efficiency losses of logic-level translation. I decided to use an FCC/CE-certified module over a bare SoC as it integrated the antenna and other essential passives.

This minimised RF layout complexity and greatly reduces RF certification costs if the device is to be commercially manufactured. The benefit of already having certain core passives also reduces the total number of unique parts in the BoM. However the limitations of this are that the total cost for a module is higher than the cost of the bare SoC and passives, and that a smaller solution footprint can be designed if a module is not used.

3.3.4 - Auxiliary Sensor Selection

To add environmental telemetry, a low-power, low-cost **Sensirion SHT30** temperature and humidity sensor was integrated onto a shared I²C bus. Importantly, both the VL53L4CD and the SHT30 share pin-compatible footprints with other sensors in their respective manufacturer portfolios. This design choice allows the platform's hardware configuration to be modified for alternative use cases (e.g., extended ranging depth or temperature accuracy) purely via a component swap and a firmware update, without requiring a PCB redesign.

3.4 - Firmware research

3.4.1 - Operating System Selection

To match the requirements of sensor monitoring and multi-protocol wireless communication, a real-time operating system (RTOS) was chosen over a traditional bare-metal super-loop architecture. While a bare metal approach minimises overhead, executing network tasks alongside sensor interactions and NVM events introduces timing risks and blocking conditions.

Zephyr RTOS was chosen as the firmware platform as it features industry-grade networking stacks that are natively supported by nRF Connect SDK in the VS Code IDE. Zephyr is the standard for firmware for Nordic Semiconductor based processors hence there is a wealth of examples available for users to adapt for their application firmware.

3.4.2 - Sensor API and Power Architecture

The implementation required selecting from one of two manufacturer-provided APIs for the VL53L4CD sensor.

- **STSW-IMG026 (Ranging Driver)** - Provides comprehensive distance data over an I²C bus. Relies on processor polling, which prevents the processor from entering low-power sleep states for extended periods, leading to increased baseline current consumption.
- **STSW-IMG034 (Presence Driver)** - Functions autonomously by executing threshold checks directly on the sensor. Operates on an interrupt-driven basis, asserting a hardware interrupt pin only when a target crosses a configured detection threshold. An I²C write is still required to deassert the interrupt pin.

To maximise battery longevity, the **STSW-IMG034** API was selected. This choice permitted a highly efficient, event-driven firmware architecture. The nRF54L15 remains in an ultra-low-power "System ON" sleep state, consuming 2.9 μ A of current (more than the 0.7 μ A floor due to full memory retention), and only wakes to process data when triggered by the sensor's hardware interrupt.

3.5 - Reference Firmware and Development Methodology

3.5.1 - Reference Firmware Architecture

To demonstrate the platform's capabilities and provide a foundation for custom development, a reference firmware implementation was developed for unidirectional object counting with environmental logging.

The application utilises an event-driven architecture to capture, queue and secure data:

- **Autonomous Presence Interrupts** - The VL53L4CD is configured using the STSW-IMG034 API, allowing the nRF54L15 to remain in ultra-low-power 'System ON' sleep state and wake up solely to process events or perform periodic logging (if configured).
- **Buffered Storage Architecture** - To ensure data persistence across power faults, events are queued in a RAM ring buffer and flushed to a 400 KB NVM partition periodically. Storing events as 32-bit timestamps, providing a theoretical capacity of over 50,000 entries (the timestamps are not strictly 32-bit due to filesystem overhead). Because the nRF54L15 utilises Resistive RAM (RRAM) for NVM, wear levelling is not necessary.
- **Periodic Environmental Telemetry** - The SHT30 temperature and humidity sensor is managed via a low-frequency polling schedule on the shared I²C bus. Its logs are directly committed to the NVM.

This reference design serves as a functional validation example that showcases all of the primary hardware subsystems, including the BLE stack, NVM storage, I²C bus, and low-power hardware interrupts under realistic constraints.

3.5.2 - Hardware / Firmware Co-design Methodology

To minimise project schedule risks and ensure rapid hardware bring-up, a hardware/software co-design methodology was employed. Firmware development was decoupled from physical PCB fabrication by implementing the firmware concurrently on an off-the-shelf Nordic Semiconductor development kit (DK). The development kit contains a SEGGER J-Link OB debugger that programs the nRF54L15 SoC on the DK and can be connected by a cable to program a custom PCB. It also supports breakpoint debugging allowing for thread activity and stack usage to be analysed.

This parallel workflow allowed the BLE stack, low-power state transitions, and sensor drivers to be fully validated and debugged in advance. Consequently, when the PCB prototype was assembled, a usable firmware version could be almost immediately flashed, reducing integration debugging to hardware-specific routing and definitions.

4 - Design and Build

4.1 - Schematic Design

Unlike the physical PCB layout stage, where Electronic Design Automation software actively flags issues, the schematic entry phase relies entirely on manual design verification. Any electrical routing or footprint mapping errors introduced here are silent and pass automated checks unnoticed.

This risk materialised during the initial design cycle: a schematic symbol adapted for the PMOS-based reverse polarity protection circuit featured non-standard pin numbering that did not match its physical footprint, resulting in a complete power delivery failure on the prototype. This highlighted the necessity of rigorous, multi-point footprint verification prior to fabrication. The development of the schematic proceeded through several distinct stages:

4.1.1 - Power Delivery and Diagnostics

The primary power stage was modelled using Texas Instruments' WEBENCH suite as an initial baseline. To elevate this reference design for custom debugging and Design for Testability (DFT), several additions were integrated as illustrated in **Fig 4.1.1.1**.

- A **PMOS-based reverse polarity protection** circuit to safeguard the board during battery swapping.
- A series **shunt resistor** for optional, high-precision power measurements.
- Dedicated diagnostic **test points** for the battery voltage (+BATT), system voltage (+2V8), ground (GND), and the shunt's high/low nodes.
- Zero-ohm current limit configuration resistor (instead of a direct connection to ground) to allow adjustment of the current limit if necessary.

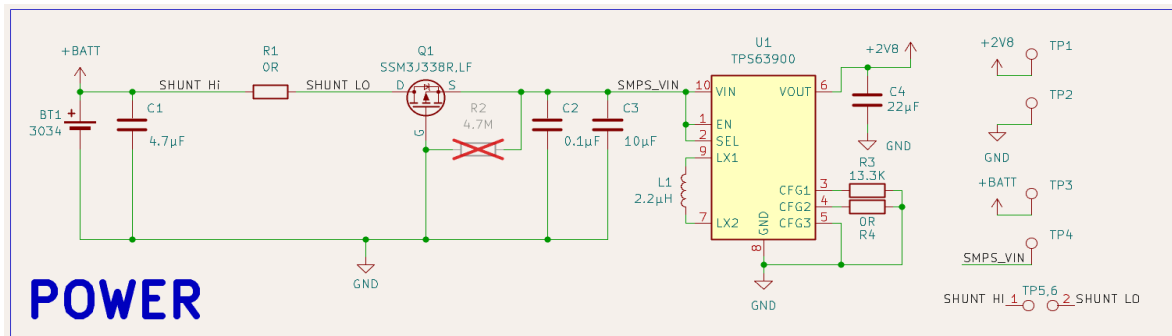


Fig 4.1.1.1 - Excerpt of schematic showing power section

4.1.2 - SoC Integration and Custom Footprint Generation

At this point the **u-blox NORA-B206** module was selected because it incorporates the highest-memory variant of the nRF54L family and is available from reputable distributors. As the nRF54 family was released relatively recently, community footprint libraries were practically non-existent.

Hence, a custom schematic symbol and high-density **Land Grid Array (LGA)** footprint had to be drawn completely from scratch, cross-referencing the manufacturer's datasheet dimensions. Using this validated symbol, the module's general-purpose I/O pins were mapped to route the shared I²C bus for the sensors, and the GPIO (GPIO1 and XSHUT) pins of the VL53L4CD could be connected.

4.1.3 - Connectorless Programming Interface

To optimise the **Bill of Materials (BoM)**, a component-free programming header footprint was selected. Rather than soldering a physical, multi-pin header to the board, the programming interface consists of exposed gold pogo-pin target pads aligned with Non-Plated Through Holes (NPTH). This mechanical choice eliminated an assembly component, saving physical board space and reducing unit weight, cost, and complexity.

4.2 - PCB Design

4.2.1 - Stackup and Mechanical Constraints

The transition of the schematic to a routing layout introduced substantial geometric and signal integrity constraints. To manage the dense routing of the high-pin-count LGA package of the SoC and maintain signal integrity, a four-layer board stackup was chosen:

- **Layer 1 (Top)** - Component placement, signal routing and some power routing.
- **Layer 2 (Inner)** - Solid ground plane for low impedance return paths.
- **Layer 3 (Inner)** - Power plane including very minimal signal routing.
- **Layer 4 (Bottom)** - Large negative battery contact pad and mechanical ground.

To reduce fabrication costs, **Via-in-Pad Plated Over (VIPPO)** technology was avoided. However, the physical dimensions of the board were strictly bound by the battery holder footprint. The selected solution, a low-cost, minimal-footprint metal retainer style holder, solders directly to the PCB to connect the positive terminal, while the negative cell terminal is pressed directly against a large exposed gold-plated (ENIG) pad on the bottom layer. Because of this large contact area, trace routing on the bottom layer was highly restricted.

4.2.2 - Battery Interface Failure Analysis and Mitigation

Due to cost-saving measures during PCB manufacturing, vias routed through the bottom-layer battery contact zone were tented with solder mask rather than being plugged/filled with epoxy. Because tented vias are physically raised and less durable than filled vias, they proved highly vulnerable to mechanical stress.

Fig 4.2.2.1 - Rear PCB view without holder

During early testing, the sharp metallic edge of the coin-cell battery casing abraded the PCB surface during insertion. After approximately 100 battery insertions across two prototypes, this continuous mechanical abrasion scraped away the thin solder mask, causing the 2.8 V power net to short to ground whenever the battery was installed.

To resolve this issue on the immediate hardware prototypes, a manual retrofit was applied by reflowing a bead of solder directly onto the centre ground contact. This localised solder dome physically elevated the battery case by roughly a millimetre. This clearance successfully lifted the sharp outer edge of the battery casing away from the vulnerable 2.8 V vias, minimising future abrasion and temporarily restoring function to the broken prototypes.



To translate this physical workaround into a clean, repeatable manufacturing step, the solder and paste masks were revised to create **PCB v1.12**, from which I later designed v1.20. As shown in **Figure 4.2.2.1**, three dedicated, solderable spacer pads (the grey pill-shaped landing pads) were integrated directly into the central gold contact zone. During assembly, if necessary, solder can be cleanly reflowed onto these pads to create precise, solid mechanical shims that raise the cell safely above the surrounding board plane.

4.2.3 - Layout Optimisation and Noise Mitigation

The top-layer layout was optimised to balance noise immunity and signal integrity. To suppress electromagnetic interference from the Switch-Mode Power Supply (SMPS), switching loop areas were minimised. Similarly, I²C traces were routed short and direct to minimise parasitic bus capacitance, keeping rise times fast enough to run the bus reliably at 100 kHz using only the weak internal SoC pull-ups. Finally, copper pours and ground stitching vias minimised trace resistance and were used to reduce return path impedance.

4.3 - PCB Assembly

The high-density layout made hand soldering hard / impossible due to 0402 (1.0 mm x 0.5 mm) passives and the u-blox NORA-B206 LGA package, which features hidden pads with a 0.9 mm pitch. Consequently, a surface-mount reflow process was utilised, employing an electropolished stencil, 137°C bismuth-based solder paste, and a laboratory hot plate. A custom alignment jig was created to overcome initial deposition inconsistencies, securing a relatively repeatable (for my use case) paste-application success rate over 70%. To ensure hardware longevity and supply-chain resilience, all components were sourced from reputable distributors, filtering out any Not Recommended for New Designs (NRND) or obsolete parts.

4.4 - Firmware Design

The firmware was developed in C using the **Zephyr RTOS** within the Nordic **nRF Connect SDK**, compiling to approximately 140 kB of NVM and 55 kB of RAM. The architecture decouples the main state machine loop in `main.c` from peripheral drivers managed via modular service headers.

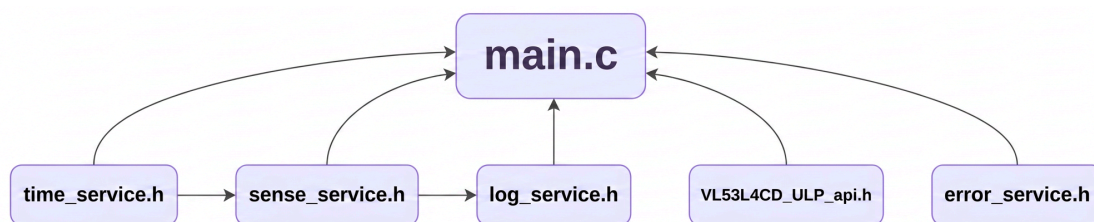


Fig 4.4.0.1 – Modular firmware structure diagram, arrows indicate consumed service headers

4.4.1 - BLE Profile and Custom GATT Service

To optimise power consumption, the device functions as a **Bluetooth Low Energy (BLE) GATT Server** that limits advertising to two minutes post-reset or after covering the sensor for 9.6 seconds (configurable via Kconfig flags). The custom GATT service exposes encrypted read characteristics (`count`, `temperature`, `humidity`, `timestamp`), write characteristics (`base_timestamp` for time synchronisation, `command` for OTA filesystem control), and a notify characteristic (`download`) to stream logged data using a custom protocol.

4.4.2 - State Machine and Adaptive Ranging

The core system loop is governed by a state machine optimised for minimal CPU active time. The device enters a low-power sleep state during user-defined "inactive hours" to conserve energy. When active, the sensor ranging rate dynamically scales from a standby 2 Hz (500 ms interval) to an active 10 Hz (100 ms interval) upon detecting an object, capturing high-density events for a programmable duration before returning to standby. An optional debouncing filter can also be enabled to require the target to remain in the ToF's field-of-view for more than one detection. This helps increase consistency near the configured detection range at the expense of reducing the effective ranging rate by a factor of the minimum number of consecutive detections for a valid reading, n .

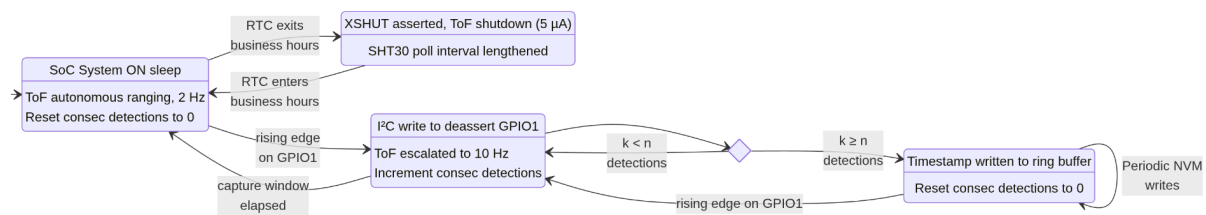


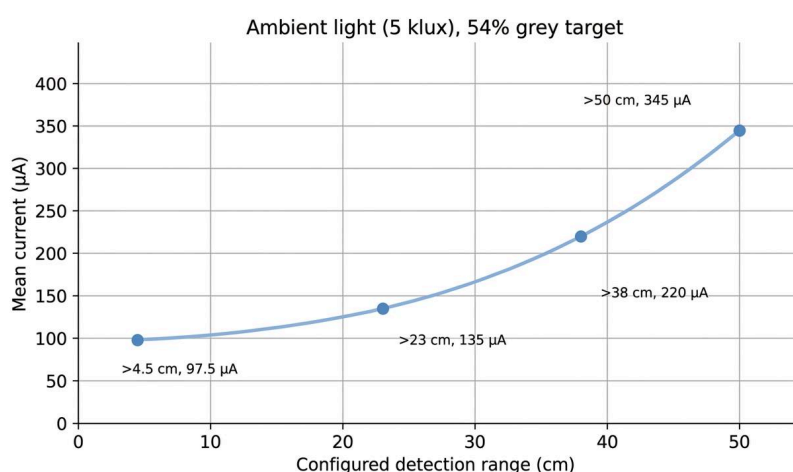
Fig 4.4.2.1 – System state machine. k = consecutive detection count; n = user-configured debounce threshold ($n = 1$ disables filtering)

5 - Evaluation of Final Product

To assess the physical viability of the prototype, the assembled hardware was validated and evaluated against the key technical and design constraints defined in the initial specification.

5.1 - Performance Testing

Validation focused on characterising the physical trade-offs of the Time-of-Flight (ToF) sensor. Under default firmware settings, the system achieves an effective detection range of 600 mm for a white paper target (in ambient light). Because the VL53L4CD allows for dynamic parameter tuning, the active current draw scales directly with the requested ranging distance and frequency. Adjusting these parameters allows the system to trade range for reduced current consumption, as characterised in the curve below, which displays the current consumption of the ToF sensor (not the whole system).



During high-volume data transfer testing, a firmware limitation was identified: the notification based GATT characteristic struggled with packet delivery during larger bulk downloads ($m > 150$ entries). This throughput issue does not affect real-time object counting or logging. However, it limits retrieval of historical timestamp records over BLE.

As an application-layer issue rather than a hardware fault, this is readily addressable given the platform's open, user-programmable design (Spec 3.1), but remains a target for development.

Fig 5.1.0.1 - VL53L4CD current consumption against range (in isolation)

Spec #	Metric	Target Specification	Achieved Performance
2.1	Low cost	Total cost per unit under £20.	Over 1ku, cost per unit near £6-£7.
2.2	Form factor	Solution footprint area under 1000 mm ² .	PCB measuring 22.1 mm x 30.2 mm, area approx. 667 mm ² .
2.3	Low power	At least 4 weeks of battery life in continuous operation.	Up to 8 weeks of battery life facilitated by current consumption in the μ A range*.
2.4 / 2.5	Connectivity	Wireless data download up to 10 m (line-of-sight).	Integrated 2.4 GHz radio, line of sight Bluetooth connectivity up to 24 m.
2.6	Accuracy	Detect at least 90% of the passes of a white target at 0.5m.	97% accuracy (white target at 0.5 m) , using default shipped firmware.
3.1 / 3.2	Usability	Documented open-source repository.	100% open-source design files [3], including a quickstart guide, labelled renders and multiple README files.

Fig 5.1.0.2 - Requirement Traceability Table

**Shipped firmware current consumption: 470 μ A during active hours, ~60 μ A during sleep hours. Measured via the §4.1.1 shunt. 150 μ A in active hours is achievable with a lowered ranging rate and maximum distance to achieve an estimated 8 weeks of continuous battery-powered operation.*

6 - Recommendations for Future Work

First, the download section of the example firmware should be rewritten to ensure a complete timestamping example. Secondly, higher resistance pull-up resistors for the VL53L4CD's GPIO pins should be tested to bring down sleep current closer to a theoretical lower bound of ~25 μ A. In the long term, a larger version could be developed to facilitate a rechargeable cell with USB-C charging and an ultrasonic ToF sensor. This would improve detection consistency for optically difficult targets and allow the use of larger cells.

7 - Acknowledgements

I would like to thank the Royal Air Force for funding my Arkwright Scholarship which made this project possible, The Smallpeice Trust for administering my Arkwright Scholarship, and PCBWay for reaching out to sponsor free PCB fabrication for the project.

8 - References

[1] STMicroelectronics, "VL53L4CD: Time-of-Flight high accuracy proximity sensor," Datasheet, 2021. <https://www.st.com/resource/en/datasheet/vl53l4cd.pdf>

[2] Nordic Semiconductor, "Current consumption," nRF54L15 Product Specification. https://docs.nordicsemi.com/r/bundle/ps_nrf54l15/page/chapters/current_consumption/doc/current_consumption

[3] A. Zembrzuski, "AZSensorSuite GitHub Repository," 2026. <https://github.com/AdamZembrzuski/SensorSuite>